Topic 1- Motion (Kinematics)

Q1-1: What is the difference between "Distance" and "Displacement"? What is the difference between "Speed" and "Velocity"? Please take examples.

- Reference Video: `Speed and velocity | Middle school physics`
- Explained by:
- Explanation:

Q1-2: How do we define "Acceleration"? What does it mean if acceleration is negative?

- Reference Video: `Acceleration | Middle school physics`
- Explained by:
- Explanation:

Q1-3: A dog runs 10 meters East, then turns around and runs 4 meters West. What is its total Distance? What is its total Displacement?

- Reference Video: `Speed and velocity | Middle school physics`
- Explained by:
- Explanation:

Q1-4: A bicycle goes from 0 m/s to 12 m/s in 3 seconds. What is its average acceleration?

- Reference Video: `Acceleration | Middle school physics`
- Explained by
- Explanation:

Q1-5: A person walks in a straight line. From time t = 0 s to t = 4 s, their position (x) goes from 2 meters to 10 meters.
(1) What is their velocity?
(2) If their position stays at 10 meters from t = 4 s to t = 7 s, what is their velocity during this time?

- Reference Video: `Interpreting motion data | AP Physics`
- Explained by:
- Explanation:

Q1-6: What is the difference between "Average velocity" and "Instantaneous velocity"?

- Reference Video: `Position, velocity, and speed | AP Physics`
- Explained by:
- Explanation:

Q1-7: A car starts from 0 m/s and speeds up steadily. After 5 seconds, its velocity reaches 20 m/s.
(1) Draw a quick Velocity-Time (v-t) graph for this motion.
(2) What is the total "Displacement" of the car during these 5 seconds?

- Reference Video: `Interpreting motion data | AP Physics`
- Explained by:
- Explanation:

Q1-8: If a car starts from rest ($v_0 = 0$) and moves with a constant acceleration of 3 m/s^2, what is its velocity after 5 seconds?

- Reference Video: `Developing kinematic equations from data | AP Physics`
- Explained by:
- Explanation:

Topic 2 : Forces & Newton's Laws

Q2-1: You are riding on a bus, and it suddenly hits the brakes. Your body lunges forward. Using Newton's First Law (Law of Inertia), explain *why* your body moves forward even though nothing pushed you from behind.

- Reference Video: `Newton's first law | Middle school physics`
- Explained by:
- Explanation:

Q2-2: Imagine a skydiver falling through the air with a parachute open, descending at a steady, constant speed. Draw a Free-Body Diagram (FBD) for the skydiver when he reaches the terminal velocity. Explain the terminal velocity first.

- Reference Video: `Forces and free-body diagrams | AP Physics`
- Explained by:
- Explanation:

Q2-3: A supermarket shopping cart full of groceries has a mass of 40 kg. If you want to accelerate the cart from a stop to 2 m/s^2 down the aisle, how much net force must you apply? Would it require more or less force to push an empty cart? Why?

- Reference Video: `Newton's second law | Middle school physics`
- Explained by:
- Explanation:

Q2-4: When a swimmer pushes off the pool wall with their feet, they shoot forward through the water. Explain Newton's Third Law and use the law to explain the phenomenon.

- Reference Video: `Newton's third law | Middle school physics`
- Explained by:
- Explanation:

Q2-5: An astronaut has a mass of 70 kg on Earth. When they travel to the Moon, they feel much lighter. Explain the “mass” and “weight” respectively and their difference. Based on the concepts of "Mass" and "Weight", does the astronaut's mass change on the Moon? Calculate the astronaut's weight (Force of gravity) on Earth. (Assume $g = 10\,\mathrm{m/s^2}$)

- Reference Video: `Weight, apparent weight, and weightlessness | AP Physics`
- Explained by:
- Explanation:

Q2-6: In a game of tug-of-war, Team A pulls a 10 kg center ring to the left with 150 N of force, while Team B pulls it to the right with 120 N of force.
(1) Draw the free body diagram of the system.
(2) What is the Net Force (magnitude and direction)?
(3) Calculate the acceleration of the center ring.

- Reference Video: `Forces and free-body diagrams | AP Physics`
- Explained by:
- Explanation:

Q2-7: You are standing on an electronic scale inside an elevator. You have a mass of 60 kg (Gravity force = 600 N). Suddenly, the elevator accelerates *upwards* in 2 m/s^2. Draw your Free-Body Diagram. What number will be shown on the scale?

- Reference Video: `Weight, apparent weight, and weightlessness | AP Physics`
- Explained by:
- Explanation:

Q2-8: A 1000 kg car is traveling on a straight road at 15 m/s. The driver sees a red light and hits the brakes, creating a constant stopping force of 3000 N in the opposite direction of motion.
(1) Draw the free-body diagram of the system.
(2) Calculate the car's deceleration.
(3) Based on your kinematic equations, how many seconds will it take for the car to completely stop?

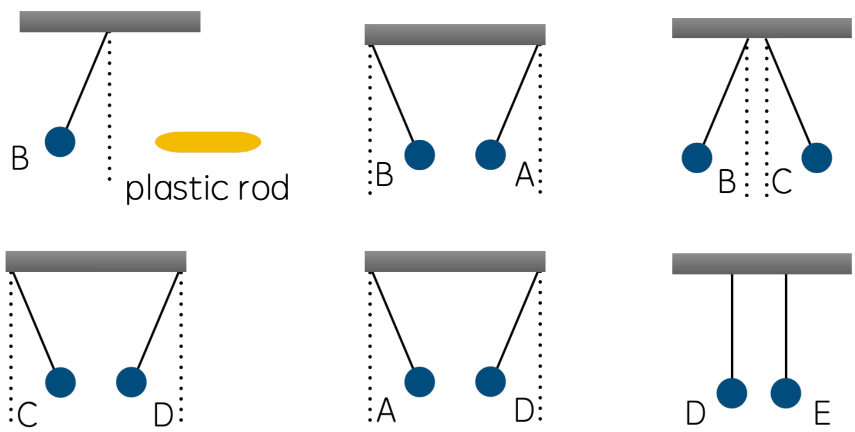
- Reference Video: `Newton's second law | Middle school physics`
- Explained by:
- Explanation:

Topic 3: Waves and Electromagnetism

Q3-1: When you take off a sweater in winter, it might stick to your shirt. Based on the concept of electric force, what type of charges (positive/negative) must the sweater and the shirt have for them to attract each other? If you pull the sweater further away, what happens to the strength of this electric force? Explain the reason by electrostatic force strength.

- Reference Video: `Electric force | Middle school physics`
- Explained by:
- Explanation:

Q3-2: Refer to the provided image . Five small conductive spheres (A, B, C, D, E) are hanging from insulating threads. A plastic rod rubbed with fur is used in the experiments. Based on the six interaction diagrams (1) through (6), deduce the electrical charge (Positive, Negative, or Neutral) of each sphere. Briefly explain your logical reasoning.



- Reference Video: `Electric force | Middle school physics`
- Explained by:
- Explanation:

Q3-3: If we consider the Earth as a giant magnet, does the Earth's Geographic North Pole and South Pole correspond to the magnetic N-pole or S-pole? Explain your answer by describing how a standard compass needle interacts with the Earth's magnetic field.

- Reference Video: `Magnetic force | Middle school physics`
- Explained by:
- Explanation:

Q3-4: Draw a standard bar magnet and sketch its magnetic field lines. Based on your drawing, state which direction the field lines flow, and explain how the electromagnetic lines indicates the strength of the magnetic force.

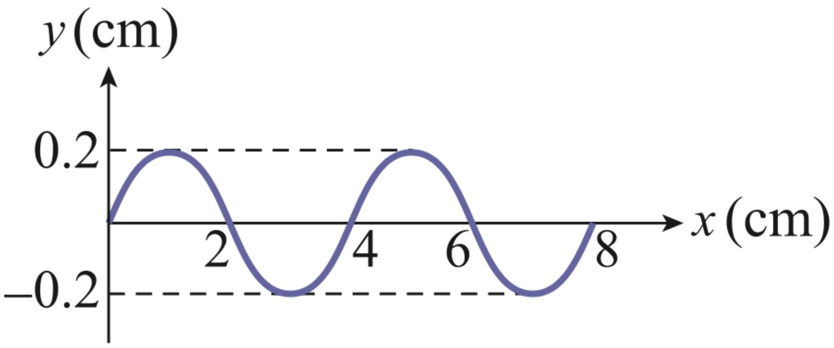
- Reference Video: `Magnetic force | Middle school physics`

- Explained by:
- Explanation:

Q3-5: As ocean waves travel toward the shore, the water becomes shallower. Surfers notice that waves further out from the shore seem to "catch up" with the waves closer to the beach, causing the wave front to steepen and form great surfable waves. Based on wave properties, how does the change in medium depth affect the wave speed, causing the wave in deeper water behind to travel faster than the wave in shallower water ahead?

- Reference Video: `Wave properties | Middle school physics`
- Explained by:
- Explanation:

Q3-6: Refer to the provided image. The graph plots a continuous transverse wave. First, identify the wave's Amplitude and Wavelength directly from the x-y graph . If it is known that the wave speed is 4 m/s, calculate the Period (T) of this wave in seconds. Show your steps.



- Reference Video: `Wave properties | Middle school physics`
- Explained by:
- Explanation:

Q3-7: A sci-fi movie describes a scene where a spaceship explodes in outer space. In the movie, you first *hear* the loud explosion, and a few seconds later, you *see* the flash of light emitted from the blast. Based on the physics of sound and light waves, point out TWO major scientific errors in this scene.

- Reference Video: `Sound waves | Middle school physics`
- Explained by:
- Explanation:

Q3-8: You are standing in front of a tall cliff and shout loudly. You hear your own echo exactly 2 seconds later. If the speed of sound in the air is approximately 340 m/s, how far away is the cliff? If you performed this same experiment underwater, would you hear the echo sooner or later? Explain with scientific reason.

- Reference Video: `Sound waves | Middle school physics`
- Explained by:
- Explanation: